

Emerging Approaches in CG Education Aimed at Enhancing Visual Communication Skills through Reverse Engineering

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ABSTRACT

The proposed research applies reverse engineering techniques in the field of computer graphics (CG) production education at Japanese vocational schools. It involves the development of a "Philosophical Observation Decomposition Table" and a "Concept Decomposition Table" to analyze the relationship between words and visuals, along with artistic elements. This methodology is designed to be both educational and enjoyable. Furthermore, the study suggests the utilization of AI technologies, such as ChatGPT, to expand the scope of CG education beyond technical skills, encompassing soft skills like communication and creativity.

CCS CONCEPTS

• **Applied computing** Education; Interactive learning environments; Education; Collaborative learning; • **Social and professional topics** Professional topics; Computing education; Model curricula;

KEYWORDS

Education-Industry Collaboration, Skill Gap, Sustainable Education Program, Diversified Needs and Skills

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1 INTRODUCTION

In the field of CG entertainment production, effectively conveying various concepts and ideas through visual media requires advanced visual literacy and communication skills. In engineering education, the successful utilization of decomposition and reconstruction as reverse engineering teaching methods. However, in the education sector focused on visual arts for CG production, there is a notable absence of widespread adoption of such methods of decomposition and reconstruction. This deficiency hampers students' ability to adequately develop visual literacy and analytical thinking skills,

posing a significant challenge. To bridge this gap, it is necessary to devise new educational methods that incorporate the techniques of reverse engineering, which have been effective in engineering education. This initiative aims to improve the skills necessary for effective communication through visual media, which have not been adequately addressed in the CG education methods presented by Balreira et al., 2017.

2 INNOVATING CG EDUCATION THROUGH OBJECTIVE AND SUBJECTIVE QUALITIES

This study is philosophically and academically grounded in Aristotle's theory of four causes, as elucidated by Hocutt, 1974 and John Locke's "An Essay Concerning Human Understanding" by Locke and Kato, 2004. Based on this theoretical framework, "primary qualities" (objective properties) and "secondary qualities" (subjective properties) within visual images are intentionally distinguished and observed. Specifically, the "Philosophical Observation Decomposition Table" is used to explore "primary qualities" in an engaging manner, encouraging instructors to integrate this with the comprehensive knowledge of CG by Alley et al., 2006. Furthermore, drawing from the theory of visual arts proposed by Bachar and Suryavanshi, 2021, subjective elements as "secondary qualities" are captured and analyzed.

Ultimately, by utilizing AI in a creative context for reconstruction, the process facilitates an awareness of new perspectives and interpretations. These sequential learning processes are interrelated and complementary. The Philosophical Observation Decomposition Table lays the foundation for understanding and conveying basic viewpoints through the differentiation between objective and subjective observations. Meanwhile, the Concept Decomposition Table provides a framework for an in-depth analysis of the artistic expression and interpretation of subjects.

3 METHODOLOGY

The study targeted 140 first-year students, mostly 18 years old (70% male), at a vocational school. These students participated in three-hour visual communication sessions, three times a week over a two-week period. The sessions were sequentially designed to cover "Philosophical Observation Decomposition Table," "Concept Decomposition Table," and "Human-AI Collaboration." These initiatives were interrelated and implemented as a comprehensive program to enhance observation and creation through visual communication. The survey, based on the Torrance Tests of Creative Thinking, were conducted online with ethical considerations in mind. This approach aimed to compare changes in perspective and communication skills within group activities against traditional curricula, such as drawing education, aimed at improving observational skills. In

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addition, expert interviews were conducted with three companies to explore the effectiveness of these initiatives for employment in the CG industry.

The methodology for the "Philosophical Observation Decomposition Table" (Appendix 1) was obtained after three sessions of 50 minutes each, shown in Appendix 2. Participants aimed to achieve heightened awareness through dialogue while alternating roles as objective analysts, subjective observers, and illustrators. The "Concept Decomposition Table" (Appendix 3) which focuses on the decomposition of artistic and cultural elements, is obtained by conducting two 75-minute sessions, shown in Appendix 4. The decomposed words were recorded on the "Worldview Creation Sheet" (Appendix 5) and reconstructed with AI assistance. Appendices 2, 4, and 5 also include case studies and survey results. Axial coding based on the grounded theory method (Strauss and Corbin, 1990) was employed for qualitative data analysis to categorize important vocabularies.

4 RESULTS AND DISCUSSION

The "Philosophical Decomposition Table" initiative revealed that 33% of participants scored 6 or higher, indicating a significant change in their awareness of image decomposition and understanding (Appendix 3). Regarding the improvement of visual communication skills, 31 participants (26.3%) gave the highest score of 7, suggesting a notable shift in awareness. For changes in awareness towards collaborative work, 44 participants (37.3%) anticipated future benefits, demonstrating that the majority gained substantial insights. These findings suggest that the "Philosophical Decomposition Table" is effective in enhancing students' observational skills, communication abilities, and awareness of collaborative work.

In the "Concept Art Decomposition" initiative, more than half of the students reported an improvement in observational skills and the discovery of new perspectives compared to conventional drawing classes (Appendix 8). Free responses regarding "Human-AI Collaboration" showed mostly positive opinions (Appendix 9). These results imply that the "Concept Art Decomposition Strategy" is effective in enhancing students' observational skills and fostering new perspectives. However, considering that not all students may experience the same effects, personalized instruction tailored to individual differences is necessary.

Interviews with industry experts revealed that the application of the reverse engineering concept aimed at enhancing visual communication skills is widely recognized as beneficial in the CG industry. The images selected by instructors significantly influence students' insights, emphasizing the importance of intentional choice in teaching materials. While technical proficiency is crucial for new graduates, industry-specific communication skills are often acquired through practical experience. However, the immediate applicability of these skills remains uncertain, necessitating a balance between technical knowledge and communication skills in educational program design. Additionally, as noted by Duit et al., 2007, educators must consider the relationship between subject matter knowledge and educational structure.

5 CONCLUSION

This study demonstrated the potential of applying reverse engineering techniques in CG production education to enhance visual

communication skills. However, longitudinal research is essential to confirm the long-term effectiveness of this educational approach. Such research can be conducted by tracking students' skill improvement and employability in the CG industry. Furthermore, the proposed methods should be applied and tested in more diverse educational settings and cultures for further refinement. Developing an integrated program that combines the proposed approach with traditional CG education is also critical.

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